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Palaeomagnetic investigation of Tertiary lava from Barrington Tops, NSW, Australia, using thermal and microwave techniques

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Abstract

Tertiary lava (53 Ma) from Barrington Tops, New South Wales, Australia has been studied using conventional thermal and microwave techniques. Twenty-seven flows in two sections were sampled and their rock magnetic characteristics determined. On heating the majority of samples exhibited a single Curie temperature at around 200°C, indicating a titanium rich titanomagnetite. Some samples exhibited two magnetic phases and a few a single low titanium titanomagnetite phase. Thermal demagnetisation yielded a mean direction of $D = 189.5$, $I = 63.6$, $\alpha_{95} = 4.3$, which corroborates the previous findings of Wellman et al. [Geophys. J. R. Astron. Soc. 18 (1969) 371–395]. A pilot conventional Thellier palaeointensity analysis was unsuccessful mainly due to the samples being highly susceptible to thermo-chemical alteration. Far greater success was achieved (58%) using the microwave palaeointensity technique, where heating of the bulk sample and hence alteration is vastly reduced. Palaeointensity estimates range from 3 to 28 μT (mean 11 ± 5 μT), which, assuming that the remanence is a primary thermal remanent magnetisation, indicates a low field intensity (a fifth of the present day value) in the early Tertiary. This study demonstrates the applicability of the microwave palaeointensity technique to ancient lava. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Australia; paleointensity; microwave methods; Tertiary

1. Introduction

Determining the intensity of the Earth's magnetic field through time is of fundamental importance for geophysics and an important constraint on models of deep Earth processes. However, large temporal gaps exist in the palaeointensity record as it is difficult to obtain reliable estimates of field intensity. Barrington Tops Volcano, New South Wales, Australia has been chosen as a study area for determining field intensities as there is not only a lack of Southern Hemisphere palaeointensity data but also a global lack of data in the early Tertiary.

The common failure of conventional palaeointensity techniques due to thermal alteration of samples during the experiment led to the development of the microwave palaeointensity technique [2,3]. High frequency microwaves directly stimulate the magnetic minerals, bypassing the need to

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heat the bulk sample. Therefore, the bulk sample experiences far less heating than in conventional experiments and, consequently, alteration is greatly reduced (e.g. [4]). The microwave/SQUID magnetometer system at Liverpool University, operating at a microwave frequency of 8.2 GHz, has been used successfully for palaeointensity studies using ceramics [5,6] and historic lava [4,7]. The samples used were all dominated by a single component of magnetisation. To extend the use of the microwave palaeointensity technique, the isolation of a primary component of magnetisation in samples with more complex magnetisation histories is required. The Barrington Tops lava has been the subject of a pilot Thellier palaeointensity study and a more comprehensive microwave palaeointensity investigation. This study therefore provides an opportunity to assess the suitability of the microwave technique with ancient lava as well as provide new palaeointensity data.

2. Geological setting and sampling

2.1. Geological setting

Barrington Tops volcanic field covers ~2400 km^2 of the eastern margin of Australia and is mainly the result of relatively short-lived periods of intra-plate volcanism during the Cenozoic Era. The age of these lavas was previously estimated at 54 Ma [8], though new K–Ar dates now suggest two main episodes of volcanic activity at ~59 Ma and ~55–52 Ma, with quieter zircon-bearing eruptions at ~26 Ma, ~18 Ma and 5 Ma, based on fission track dates [9]. Lavas used in this study are thought to be ~53 Ma (sampled between the sections studied in [1]).

The stratigraphy of the area has been described by Mason [10] as a sequence of tholeiitic, olivine rich and undifferentiated basalts unconformably overlying folded carboniferous alluvium. The dominant alkaline intergranular olivine basalts were mainly sampled for this study, with flow thickness ranging from 2 to 15 m. The flows exhibit well-developed columnar jointing and show amygdales of calcite and zeolite in the flow tops. Some flow tops are strongly ferruginised.

2.2. Sampling

Sampling took place around the Prospero Trig area in 1997 (Fig. 1). A traverse of 6 km along Barrington Tops Forest Road was conducted between the sections of Semphill Creek and Stewarts Brook [1], starting at the Moonan Brook Dingo Gate (151°20′45″E, 31°56′00″S), sampling the road-cuts westwards (eastwards) for section 1 (2). This area was chosen for its overall excellent rock exposure and comprehensive stratigraphic and palaeomagnetic studies [1,10], though some thin flows were missed due to occasional poor exposure and accessibility. The section is comparable to a combination of the central and eastern road sections of Mason's stratigraphic study [10]. A petrol-powered rock drill was used to sample 25 mm diameter cores from the outcrops, with an average of seven cores being taken from each flow. Twenty-one flows were sampled from a total of 27 distinct flows observed in section 1. Six flows were sampled in section 2. Cores were oriented in the field with a sun compass, using the sun or a distant geographic landmark as a reference point. Each core was cut into three sub-samples (for thermal demagnetisation, Thellier analysis and microwave analysis). Core bottoms were also used for rock magnetic experiments.

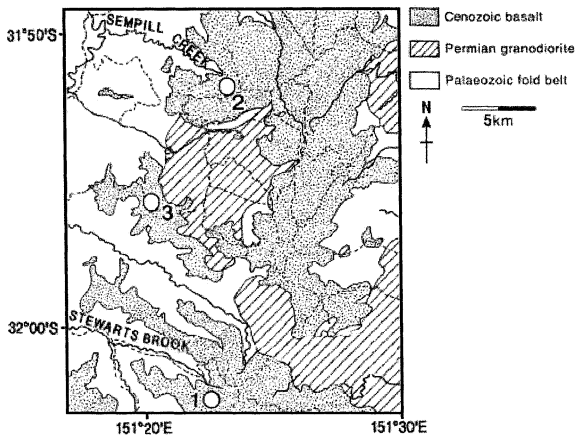


Fig. 1. Location map (redrawn from [9]) showing the Stewarts Brook (1) and Semphill Creek (2) sections [1] and Moonan Brook Dingo Gate (3) section (this study).

3. Magnetic mineralogy

Thermomagnetic, hysteresis and susceptibility (room and low temperature) experiments were carried out on two samples per flow (total 54 samples). Analyses were performed using a Curie balance and Magnetic Measurements Variable Field Translation Balance. Samples were divided into three categories (A, B and C) based mainly on their thermomagnetic properties (Fig. 2). Type A samples contain a single low temperature Curie point. Type B samples contain two discernible Curie points and type C samples are dominated by a single high temperature Curie point. Over half the samples (31) are of type A, 13 are type B and 10 are type C. All samples plot in the pseudo single domain range of a Day plot [11].

Type A samples contain a titanium rich titanomagnetite and these samples are highly susceptible to alteration with the creation of new magnetic material between 250 and 300°C. Five out of 31 samples exhibited disproportionation peaks, indicating that the titanium rich titanomagnetite has undergone low temperature oxidation. Type C samples contain a single high Curie temperature ferrimagnetic phase of a low titanium titanomag-

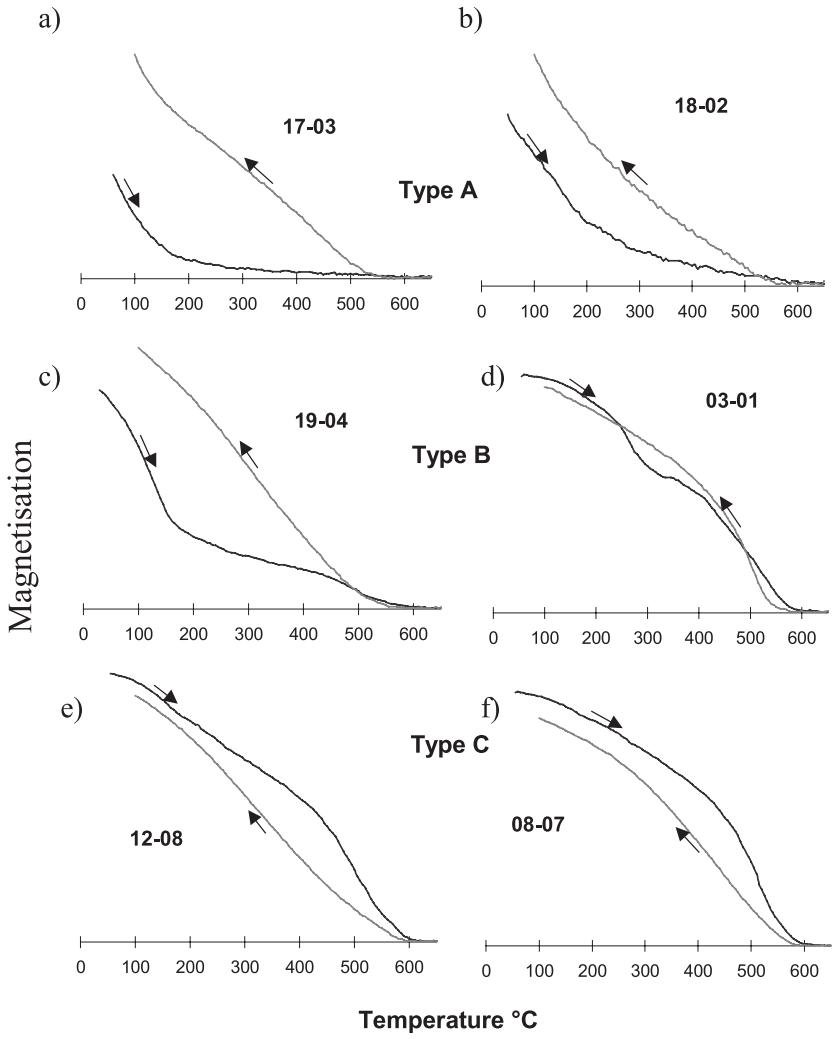


Fig. 2. Typical Curie curves; a and b are classified type A, c and d type B and e and f type C. See text for further explanation.

netite and experience less dramatic alteration on heating. Type B samples have oxidation states between A and C and contain two ferrimagnetic phases.

4. Thermal analysis

Samples were heated in a Magnetic Measurements Thermal Demagnetiser and measured using a Molspin spinner magnetometer.

4.1. Thermal demagnetisation

Thermal demagnetisation was carried out on four samples per flow for section 1 and seven samples per flow for section 2. Orthogonal vector plots (OVPs) can be seen in Fig. 3. Using principal component analysis, characteristic directions could be obtained for 16 out of the 21 flows in section 1 and one flow out of six in section 2. All flows were reversely magnetised. Table 1 gives the flow means and the overall mean direction evaluated from section 1 (it was not possible to be certain how sections 1 and 2 were related) as well as the mean direction obtained from an earlier study by Wellman et al. [1]. It can be seen that this study agrees with the earlier results where blanket AF demagnetisation was carried out to determine the primary direction of magnetisation.

It has been suggested that Barrington Volcano and other volcanoes in Eastern Australia (Liverpool and Nandewar volcanoes) give anomalous palaeopoles [12–15]. As this study agrees with the earlier study, the question regarding the validity of the palaeopole derived from Barrington Tops remains. Two reasons have been given for the possibly anomalous basalt poles: firstly, that secular variation has not been averaged out due to episodic extrusion [14] and secondly, that the magnetisation of the rocks is more complicated than previously thought [16,13]. This is still an area for debate as ferruginised layers were seen between flows in this study and the more extensive study of Wellman et al. [1] found normal and reversed flows indicating a significant time of extrusion. There is also no apparent correlation between the rock magnetic signature and characteristic direction.

4.2. Palaeointensity study

The Coe variant [17] of the Thellier technique [18] with partial thermal remanent magnetisation (pTRM) checks was carried out on two samples per flow. Heating steps were carried out from 100°C to 600°C in steps of 100°C, 50°C or 25°C. The laboratory field used was 50 μT as an initial estimate of the ancient field as the present day field in Australia is 58 μT.

It was predicted that the Thellier experiments would be unsuccessful as the thermomagnetic analysis (Fig. 2) indicated that the majority of samples were highly susceptible to alteration on heating. Non-ideal behaviour was exhibited by all samples with failure of pTRM checks and non-linear plots of natural remanent magnetisation (NRM) remaining against thermal remanent magnetisation (TRM) gained. Typical plots are shown in Fig. 4. Where linear sections do occur the slope is shallow, suggesting that the ancient field was considerably lower that 50 μT. It might be possible to improve these results with a better choice of temperature steps and laboratory field but this will not prevent alteration of the samples during heating indicated by the failure of the pTRM checks.

5. Microwave analysis

Conventional demagnetisation and remagnetisation occurs when heat, in the form of lattice vibrations (phonons), is sufficient to generate spin waves (magnons) within the individual magnetic domains. By using high frequency micro-

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Fig. 3. OVPs for six flows, comparing thermal and microwave demagnetisation with corresponding intensity plots of temperature and microwave power against normalised magnetisation. Note OVPs for microwave demagnetisation are unoriented; open (solid) symbols represent horizontal (vertical) components.

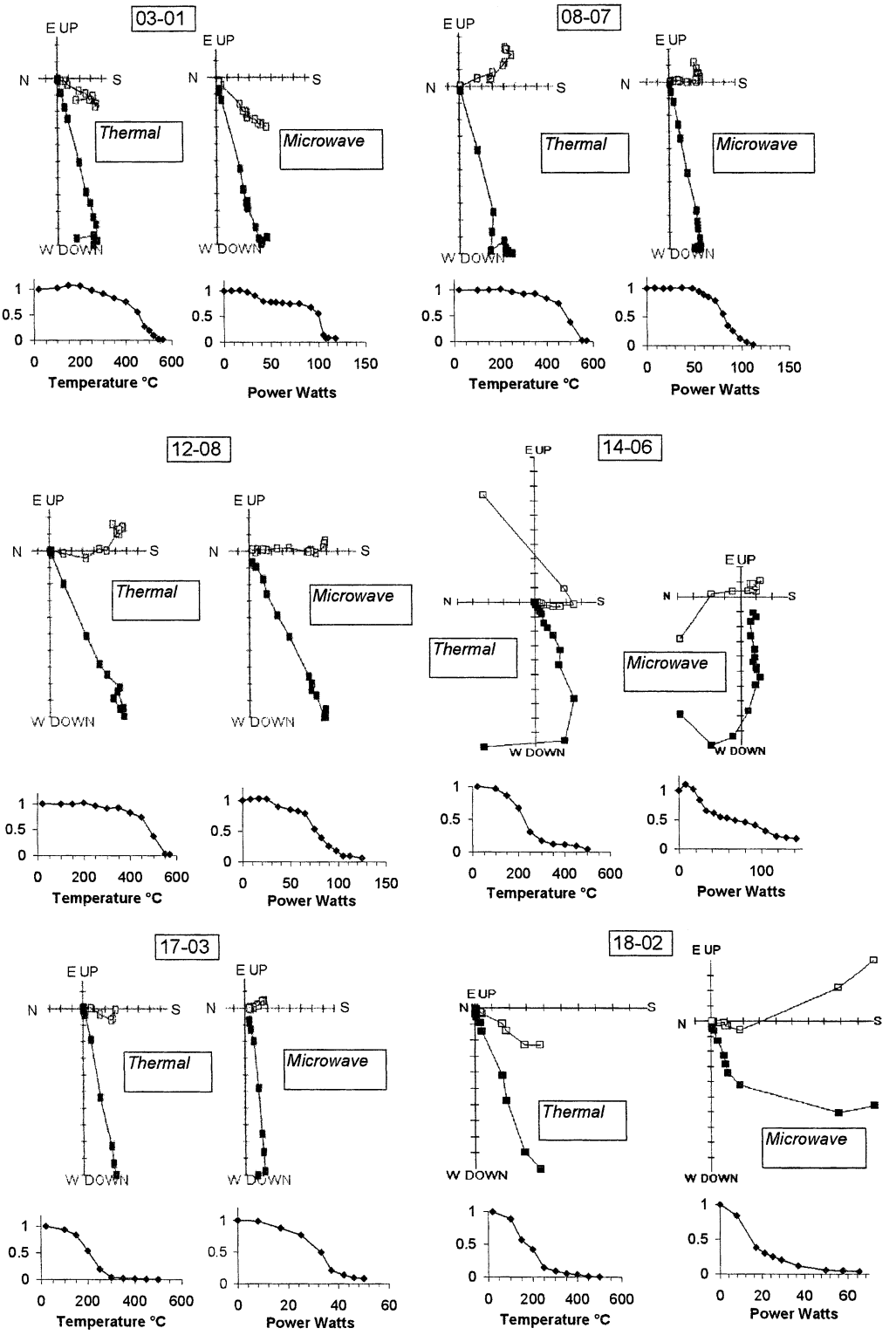

03-01
E UP
N
S
W DOWN
Thermal
Microwave
1
0.5
0
0 200 400 600
Temperature °C
0 50 100 150
Power Watts
08-07
12-08
14-06
0 100
17-03
0 20 40 60
18-02

Table 1
Flow mean directions for section 1 and section 2 and the overall mean direction from Section 1 and the study of Wellman et al. [1]

Section	Flow	n	Dec.	Inc.	α_{95}
2	R02	7	178.5	72.1	6.6
1	03	4	206.3	67.7	2.9
	05	4	163.2	67.6	9.2
	06	4	186.4	72.1	6.9
	07	4	200.9	74.5	12.9
	08	4	174.9	63.5	6.5
	09	4	199.6	57.7	9.8
	12	4	167.6	55.4	9.7
	13	4	182.4	51.7	5.5
	14	4	188.6	64.5	7.4
	15	4	190.9	52.6	6.5
	16	4	187.4	68.4	3.8
	17	4	189.6	66.0	10.4
	19	4	189.7	60.8	18.3
	18	4	205.2	65.3	5.3
	21	4	214.4	54.7	9.2
	20	4	185.9	64.7	19.6
Mean		16	189.5	63.6	4.3
Wellman et al. [1]		33	193.0	65.5	3.6

n = the number of samples. The Plat, Plong and α_{95} for section 1 are −74.2, 126.4, 5.9 compared to −70.5, 125.6, 5.3 from the Wellman et al. [1] study.

waves to directly excite the spin system the need to heat the bulk sample (to generate phonons) is bypassed [2,3]. The sample does experience some heat as the spin waves degenerate into lattice vibrations, but this is far less compared to heating the bulk sample (e.g. [7]).

Microwave energy is pumped into a cylindrical cavity tuned to the frequency of the microwaves (8.2 GHz) so that the cavity acts as an amplifier. 5 mm diameter cores of 1–3 mm length are used instead of the standard 25 mm diameter cores so that when the sample is placed in the centre of the microwave cavity the resonance characteristics of the system are not degraded significantly. Presently, it is not possible to accurately orient the mini samples, but once the sample is mounted in the microwave system it remains fixed for the duration of the experiment so that the stability of magnetisation can be monitored and OVPs obtained.

The microwaves are applied for 10 s with microwave power being increased stepwise in an analogous manner to increasing temperature settings in a thermal demagnetiser [4].

5.1. Microwave demagnetisation

Two microwave samples per standard core (one from each flow) underwent microwave demagnetisation. Representative OVP and intensity plots of microwave and thermal demagnetisation are shown in Fig. 3. Generally, similar behaviour is shown by thermal and microwave demagnetisation. However, for some samples the transition between different components of magnetisation can be more gradual with microwave demagnetisation (e.g. sample 18-02, Fig. 3). For some samples this smearing of components means that it is not possible to isolate the characteristic direction (e.g. sample 14-06, Fig. 3). The differences between microwave and thermal demagnetisation are discussed further in [19].

5.2. Microwave intensity study

The methods used with the microwave palaeointensity technique are variants of the Thellier technique. Two methods are used: the parallel field (ceramic) method [5,20] and the perpendicular field (lava) method [4,21]. The parallel field method is a double 'heating' method in which the microwave thermal remanent magnetisation (T_MRM) has the same direction (and magnitude to a best guess) as the NRM. This reduces the effects of anisotropy (hence it is used for ceramic samples which often have high anisotropy [22]). The disadvantage of this technique, being a double 'heating' technique, is that the power absorbed by the sample is required to be accurately reproduced. There is no problem accurately reproducing the amount of power that is applied to the microwave cavity but this is not the same as the amount that is absorbed by the sample as some of the power is reflected and some is absorbed by the cavity. Power absorption is affected by the resonance characteristics (Q value) of the cavity, which are sensitive to such factors as temperature and sample position. This uncertainty is overcome in the perpendicular field method in which the T_MRM is induced perpendicular to the direction

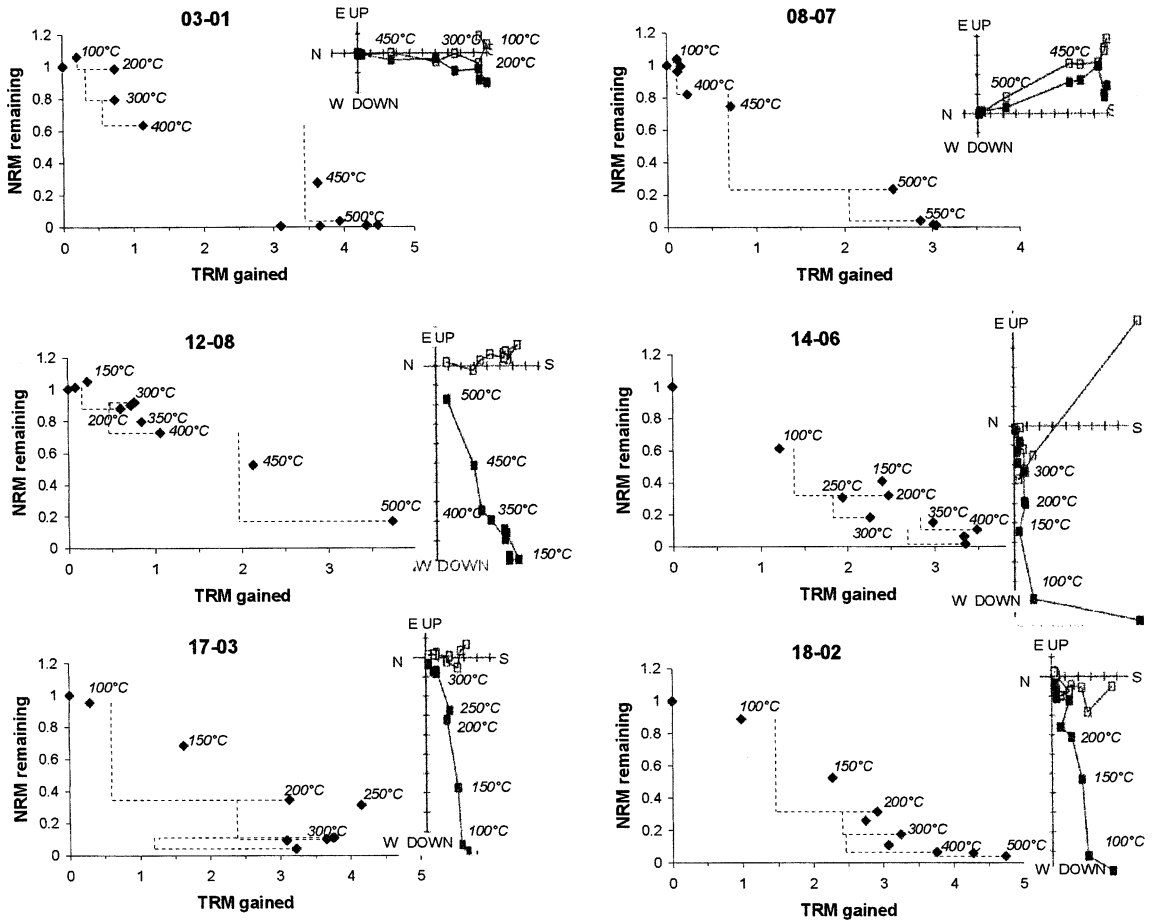


Fig. 4. Typical plots of normalised NRM remaining against TRM gained with OVPs obtained from the Thellier experiment, pTRM checks are shown by the dashed lines.

of the NRM. This method requires only one microwave application for each power step and then vector analysis is used to determine the NRM and T_MRM components. The disadvantage is that the sample has first to be demagnetised until all secondary components are removed and a stable primary direction has been isolated before starting the experiment. When determining the NRM and T_MRM components from the total vector, quality checks are in place to ensure that the two components are indeed perpendicular; if not, these data points are rejected. The analysis of results for both methods is similar to conventional Thellier experiments with data plotted on NRM remaining against T_MRM gained plots and the standard Coe et al. [23] quality parameters evaluated. PTRM checks are not normally performed during microwave experiments, because, as stated earlier, it is not possible to guarantee that the power absorbed by the sample can be exactly reproduced. In addition, a further two microwave applications at a particular power step would be needed in the perpendicular field method, thus increasing the possibility of alteration in the sample. Measuring rock magnetic properties before and after exposure to microwaves may be used as alternative alteration checks (e.g. [4]).

The approach adopted in this study was to use both microwave palaeointensity methods. Firstly, the parallel field method was carried out so that the directional behaviour could be monitored. Then, if the characteristic direction could be iso-

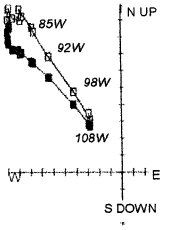
Table 2
Microwave palaeointensity results using the 8.2 GHz system

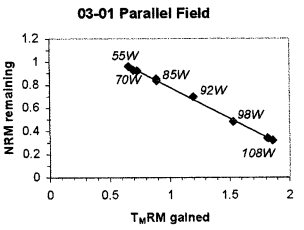
Flow	Sample	Rock mag. group	Method	n	f	g	q	F (μT)	Uncertainty
3	03-01	B	parallel	9	0.58	0.69	22.00	10.74	1.18
			perp	7	0.76	0.78	12.60	18.54	2.37
	03-02	B	parallel	6	0.46	0.67	3.20	3.69	1.21
			perp	11	0.47	0.75	15.83	5.22	0.62
5	05-03	B	parallel					no result	
	05-04	B	parallel					no result	
6	06-03	C	parallel	5	0.48	0.73	5.70	11.31	2.28
			perp	12	0.83	0.87	63.08	10.56	1.07
	06-06	C	parallel	5	0.34	0.39	3.1	17.02	5.74
			perp	11	0.73	0.75	29.70	17.58	1.86
7	07-03	C	parallel	9	0.80	0.64	16.46	15.29	1.79
			perp	11	0.70	0.81	27.32	12.51	1.33
	07-06	C	parallel	7	0.47	0.74	9.46	4.67	0.68
			perp	6	0.23	0.48	2.09	4.31	2.10
8	08-05	C	parallel	9	0.88	0.79	43.19	15.46	1.59
			perp	11	0.73	0.86	22.03	13.95	1.53
	08-07	C	parallel	8	0.76	0.79	18.50	14.33	1.63
			perp	6	0.71	0.77	18.60	14.02	1.59
9	09-02	B	parallel					no result	
	09-03	A	parallel	8	0.69	0.84	4.63	27.96	6.66
			perp	8	0.30	0.78	4.60	15.87	3.80
12	12-06	A	parallel	3	0.49	0.50	9.58	3.52	0.51
			perp	5	0.57	0.69	5.42	3.68	0.77
	12-08	C	parallel	7	0.72	0.74	24.98	9.08	0.98
			perp	15	0.63	0.86	41.00	7.04	0.72
13	13-02	C	perp	16	0.90	0.88	32.31	13.33	1.40
14	14-06a	A	parallel					no result	
15	15-01a	A	parallel					no result	
	15-06a	A	parallel					no result	
16	16-04a	A	parallel					no result	
17	17-03a	A	parallel	5	0.66	0.74	13.58	7.01	0.87
			perp	9	0.79	0.86	59.27	7.20	0.73
	17-04a	A	parallel	9	0.93	0.82	59.68	9.16	0.93
			perp	6	0.74	0.78	15.61	8.91	1.06
18	18-01a	A	parallel					no result	
	18-02a	A	parallel					no result	
19	19-01a	C	parallel					no result	

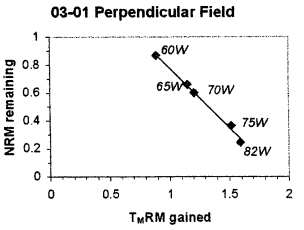
Parallel indicates the parallel field method and perp indicates the perpendicular field method, n is the number of data points, f, g and q are the quality parameters as defined by Coe et al. [23]. F is the palaeointensity estimate and the uncertainty is calculated using $1/q$ as defined by [23]. Palaeointensity estimates are accepted if there is a linear section on the NRM/T_MRM plot with $n \geq 5$ and $f \geq 0.3$; however, sample 12-06 parallel field method ($n = 3$) and sample 07-06 perpendicular field method ($f = 0.23$) have been included as the palaeointensity estimates corroborate the results obtained using the other microwave palaeointensity method.

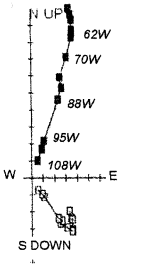
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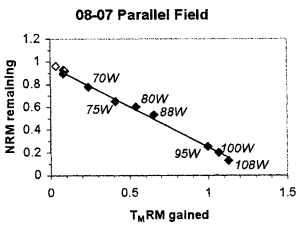
Fig. 5. Representative plots of normalised NRM remaining against T_MRM gained for microwave palaeointensity analysis using the parallel field and perpendicular field methods plus the OVP obtained during the parallel field method experiment. Closed (open) symbols represent accepted (rejected) points.

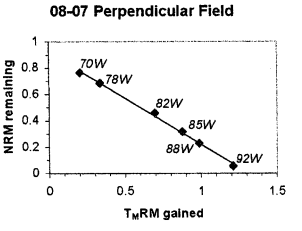

N UP
85W
92W
98W
108W
W
E
S DOWN

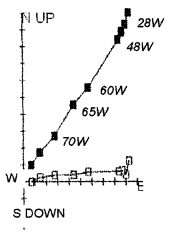

03-01 Parallel Field
NRM remaining
55W
70W
85W
92W
98W
108W
T_MRM gained

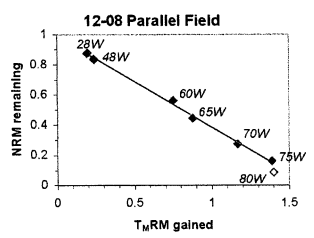

03-01 Perpendicular Field
NRM remaining
60W
65W
70W
75W
82W
T_MRM gained

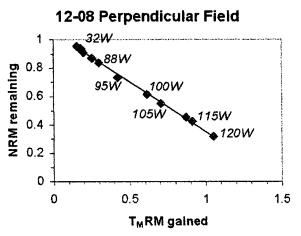

N UP
62W
70W
88W
95W
108W
W
E
S DOWN

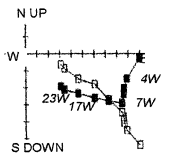

08-07 Parallel Field
NRM remaining
70W
75W
80W
88W
95W
100W
108W
T_MRM gained

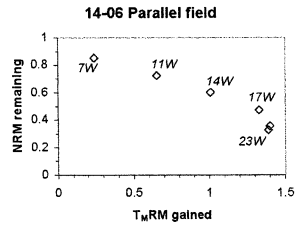

08-07 Perpendicular Field
NRM remaining
70W
78W
82W
85W
88W
92W
T_MRM gained

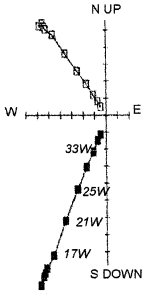

N UP
28W
48W
60W
65W
70W
W
E
S DOWN

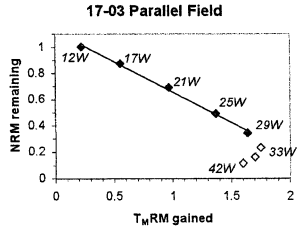

12-08 Parallel Field
NRM remaining
28W
48W
60W
65W
70W
75W
80W
T_MRM gained

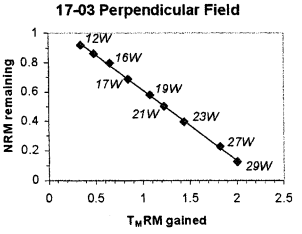

12-08 Perpendicular Field
NRM remaining
32W
88W
95W
100W
105W
115W
120W
T_MRM gained

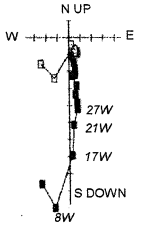

N UP
W
E
4W
23W
17W
7W
S DOWN

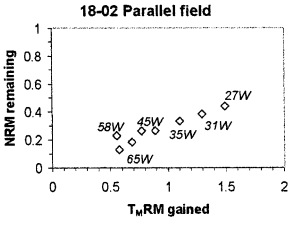

14-06 Parallel field
NRM remaining
7W
11W
14W
17W
23W
T_MRM gained


N UP
W
E
33W
25W
21W
17W
S DOWN


17-03 Parallel Field
NRM remaining
12W
17W
21W
25W
29W
33W
42W
T_MRM gained


17-03 Perpendicular Field
NRM remaining
12W
16W
17W
19W
21W
23W
27W
29W
T_MRM gained


N UP
W
E
27W
21W
17W
S DOWN
8W


18-02 Parallel field
NRM remaining
27W
58W
45W
35W
31W
65W
T_MRM gained

lated and the parallel field method yielded a linear NRM/T_MRM plot, the perpendicular field method was carried out on a sister sample from the same 25 mm diameter core. Twenty-four 25 mm cores from 14 flows were investigated. As the Thellier results suggested a low ancient field the intensity of the laboratory field used during the microwave experiments was 15 μT. Results are listed in Table 2 and representative NRM remaining against T_MRM gained plots are shown in Fig. 5. It was not possible to obtain a palaeointensity estimate for 10 out of the 24 cores. This represents a 58% success rate. Reasons for experimental failure were the inability to isolate the characteristic component of magnetisation (e.g. sample 14-06, Fig. 5) or, in some cases, no linear section was found on the NRM remaining against T_MRM gained plot. In one case, unusual behaviour was seen where the T_MRM decreased as the NRM decreased (sample 18-02, Fig. 5). This behaviour was not seen in the Thellier experiment (sample 18-02, Fig. 4).

Microwave palaeointensity estimates were obtained for samples with all three types of rock magnetic classification. As expected, type C samples were the most successful with eight out of nine determinations giving estimates. Four out of 10 type A samples and two out of five of the type B samples were successful.

The palaeointensity estimates obtained using the two microwave methods agreed, within error, for all samples apart from 03-01 and 09-03. As the parallel field method uses double the number of microwave applications compared to the perpendicular field method, it is expected that if alteration during the experiment were occurring then it would be more pronounced in the parallel field experiments. The consistency of results from the two methods is therefore an indication that alteration is not occurring. Reasons for inconsistency between the two methods could be due to alteration, sample anisotropy, incomplete removal of the secondary components of magnetisation, or small-scale inhomogeneity within the 25 mm core. Without further experimentation it is not possible to conclude which of these is the cause.

Microwave determinations from two standard cores (four microwave palaeointensity determinations) have been obtained for five flows (03, 06, 07, 12, 17). Large differences in the palaeointensity estimates can be seen for two flows (03 and 07). Detailed investigations of single lava flows [7,24,25] have found similar scale variations. The cause and meaning of the variations are still not clearly understood but the best strategy appears to be to take a large number of samples per flow and to average the results [26]. With this in mind, to gain greater confidence that the results of this study are indeed true representations of the geomagnetic field further palaeointensity analyses per flow should be performed.

The palaeointensity estimates obtained using the microwave palaeointensity technique agree with the indications from the Thellier experiments that the palaeointensity is low, with individual values ranging from 3 to 28 μT. The mean palaeointensity from nine flows is 11 ± 5 μT. This corresponds to about a fifth of the present day field in Australia.

6. Discussion and conclusions

To compare the results obtained from Barrington Tops with palaeointensity data from other geographical locations, a virtual dipole moment (VDM) has been calculated. Using the mean palaeo-inclination, the VDM is $1.80 \pm 0.82 \times 10^{22}$ Am^2. This is significantly lower than the average dipole moment for 300–0.3 Ma reported by [27] of around 5×10^{22} Am^2. Both these values have been plotted in Fig. 6 along with data in the PINT2000 database [28] and from [27] for the time period 60–40 Ma. There are only four studies [29–32] in the database for 60–40 Ma and the dipole moments show considerable scatter. It should be noted, however, that none of these studies used the Thellier method with pTRM checks, which is considered to be the most reliable method. The two data points from [27] obtained from submarine basaltic glass and using the Thellier method with pTRM checks are close to the average dipole moment. Further palaeointensity studies using material of age 60–40 Ma is needed to fully determine the behaviour of the dipole moment at this time.

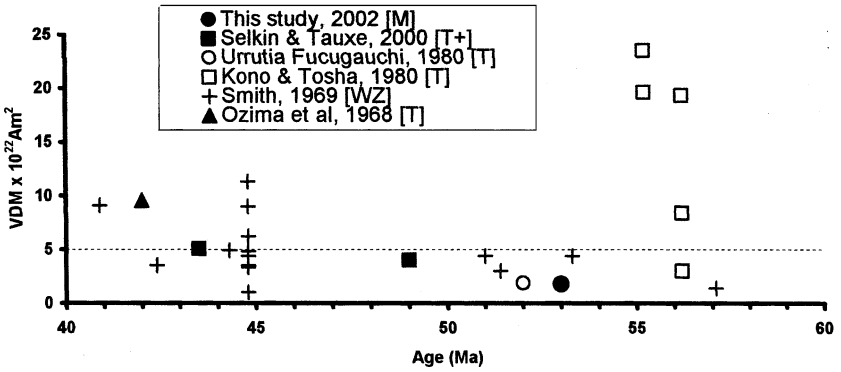


Fig. 6. VDM data using the PINT00 database [29–32], [27] and this study. Dashed line indicates the average VDM for 300–0.3 Ma as calculated by [27]. Letters in brackets indicate method used: [T] = Thellier without pTRM checks, [T+] = Thellier with pTRM checks, [M] = microwave, [W] = Wilson, [Z] = van Zijl.

This study has demonstrated that the microwave palaeointensity technique is applicable to ancient rock. In some cases, where smearing between the different components of magnetisation occurs and the primary direction of magnetisation cannot be obtained, it is not possible to obtain a palaeointensity estimate. Despite the discouraging results of the Thellier study, a 58% success rate was obtained using the microwave palaeointensity technique.

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